

Demonstrations are weights you write at inference time

A frozen model adapts to an unseen task by adding rank-one updates to a fixed-size associative state, so changing only the demonstrations changes the answer. The non-negativity that makes that state readable is the same property that caps what it can hold, at a rate of $1/\pi$. The first half is reproducible in under a minute; the second is measured.

Test-time adaptation: optimization versus context · [demonstrations-are-weights.vercel.app](#) · [github.com/snivellus38/DataForge](#) · a 131K-parameter BDH runs live in the browser, the 8M-parameter one was trained here, and BDH-CQ figures are replayed from [arXiv:2608.09888](#).

DESIGN PRESSURE

A model meeting an unseen task has two routes. It can *optimize*: turn the demonstrations into training samples and take a backward pass first. HRM ([arXiv:2506.21734](#)) and TRM ([arXiv:2510.04871](#)) do this on ARC, giving each augmented puzzle a learned identity embedding and voting over augmentations, so a hidden task cannot be scored until parameters move, at ARC Prize's \$1.48 and \$1.76 per task. Or it can *adapt in state*, writing them into a recurrent memory and answering with every parameter untouched, which needs a memory that does not grow with what it has read.

MECHANISM

Attention without a softmax and with $Q = K$ reassociates exactly into a recurrence over a state of fixed size:

$$\sigma_t = \sigma_{t-1} + \text{rope}(K)_t \otimes V_t$$
$$\text{out}_t = \sigma_t^\top \text{rope}(Q)_t$$

Each token adds a rank-one outer product to a matrix later tokens read through, so a demonstration is literally a weight. The reassociation is exact: the reference implementation ([pathwaycom/bdh](#), MIT) run both ways agrees to 2.8×10^{-14} in float64, and our 8M model's recurrent decode matches its parallel forward to 6.0×10^{-7} .

Of BDH's three unusual choices only one buys that fixed state. A softmax breaks the equivalence (1.2×10^2 ; $Q \neq K$ (1.7×10^{-5}) and dropping the ReLU (1.6×10^{-5}) do not. Softmax-freeness buys constant memory; $Q = K$ and the ReLU buy readability.

WHAT NON-NEGATIVITY BUYS

Sparse, positive, monosemantic structure is the Dragon Hatchling paper's claim ([arXiv:2509.26507](#)); the price below is ours. On our 8M model **4.93%** of units clear 1% of their peak and only **0.97%** survive the gate $y = \text{ReLU}(D_y a) \odot x$ into the residual stream. BDH is weight-tied, one neuron population and the same matrices at every iteration, so $G^* = D_x^\top E^\top$ is a standing synapse graph computed from the weights alone. Isolation in it predicts which neurons never fire on real French, at Matthews correlation **+0.940** (+1 perfect, 0 chance), against -0.001 for a degree-shuffled null over 200 draws and $+0.908$ on an independently trained Portuguese sibling.

WHAT IT COSTS

Write k rank-one bindings into a fixed $N \times D$ state and read them back: 100% at $k = 8$, 94% at $k = 32$, 48% at $k = 128$. A counterfactual isolates the cause: with *signed* keys the same state holds 384 bindings at 99%, past its rank. Mean pairwise cosine is **+0.319** for rectified keys against -0.000 for signed, and $+0.319$ is analytic: $1/\pi$ for rectified Gaussians. Non-negative vectors cannot be near-orthogonal, so they interfere: readability is paid for here. The ceiling belongs to the additive write rather than to BDH: DeltaNet ([arXiv:2406.06484](#)) replaces "the additive update in linear transformers" with a delta rule and reports it better at associative recall; TTT layers ([arXiv:2407.04620](#)) recast it as self-supervised learning.

	Decoder-only Transformer	BDH, 8M, trained and run here	BDH-CQ (replayed, not run)
Memory per token	4,608 B, unbounded	28.3 MB, flat ; cheaper past $T^* = N_{\text{total}}/2 = 6,144$ B	constant, dimensions undisclosed
Interpretability	attention maps, data-dependent	4.93% units active ; G^* predicts silence, MCC +0.940 against a -0.001 null	none published
Accuracy in domain, bits/byte	0.974 (BLOOM-560m), 1.245 (SmolLM2-135M), 1.693 (GPT-2)	0.904 at 8.0M parameters	29.5% pass@2 on ARC-AGI-1, developer-reported
Deployability	weights public	weights, code, exports and 236 checks, MIT	no weights, no API, access on request
Inference cost	2.4 MB of cache at its 512 B context	28.3 MB of state, 12.0$\times$ larger	≈ 0.85 H200 GPU-s/task, computed not billed
Accuracy out of domain	1.266 (BLOOM-560m)	2.876, a $3.2\times$ gap where the others sit at 1.28–1.36 $\times$	not reported

Above the second rule BDH wins; below it, it loses. The public models are reference points and not controls: other people, other data. Bits per byte is tokenizer-agnostic cross-entropy over identical text.

BDH AND BDH-CQ

BDH is the substrate: MIT-licensed, runnable, and what every figure above was measured on. BDH-CQ is evidence only: no public weights, no public API, and dimensions and update rules that "remain proprietary" ([arXiv:2608.09888](#), §3.3). The bridge needs no access. BDH-CQ's memory update is Eq. (1), $S_t = U_\theta(S_{t-1}, D_t)$, and §3.2 names the special case $S_t = S_{t-1} + U_\theta(D_t)$, "linear attention being the conceptually simplest standalone realization", which is what the public BDH code computes. Additive accumulation is therefore a named special case, not BDH-CQ's rule, and presenting it as the mechanism is an error.

Two replayed results. In Table 3, one demonstration added at the target complexity moves length-8 ordering from 0/24 to 13/24 exact outputs, test inputs byte-identical; in §6.3 a colour permutation defined only through demonstrations is applied to 96/96 held-out outputs.

EVIDENCE AND MATURITY

The grounding evaluation is ARC-AGI-1: 29.5% pass@2, 118 of 400 public tasks with two attempts allowed, at about 0.85 H200 GPU-seconds per task (Table 1). Five qualifications from the report. It is a pipeline score: input transformations, candidate construction and ranking included. The \$0.00070 per task is

computed at an assumed \$3 per H200-hour, not billed, and §6.6 reports the same 118 of 400 at **\$0.00265246**, $3.8\times$ higher and unreconciled, at which "57 $\times$ cheaper" becomes $15\times$. The Wilson interval [25.24, 34.15] overlaps GPT-5.6 Luna Low's higher 34.2%. The audit reproducing 29.5% was run by co-authors: developer-reported, and not an external reproduction. ConceptARC sits in the training mixture (§4.2) before being evaluated at 59.38%.

Maturity, as facts rather than a score: the architecture is public and reproducible under MIT, the system is not; there is no external reproduction and no ARC Prize leaderboard entry; Transformer-like scaling from 1B to 600B parameters is two sentences in §9.2 with no supporting data; and the public ecosystem is one third-party reimplementations with no weights.

THE OPEN QUESTION

The $1/\pi$ ceiling is a property of the additive write. BDH-CQ's rule is the general U_θ and is proprietary, and DeltaNet shows a non-additive update recalls better, so whether BDH-CQ escapes the ceiling its own named special case obeys cannot be checked from outside Pathway. That is the missing evidence. Ours sits beside it: no parameter-matched Transformer was trained as a control, and this 8M model shows no gain on repeated random spans, so demonstrations are weights in a model trained to use them that way.